Survey Methods
4.3 - Beyond questionnaires

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Outline

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- Integrating self-report with biomarkers, direct measurements, observation

Why?

Self Report Measures

Anthropometric measurements

Biospecimen

Direct observation

Low-reliability

Better measurement

	Number of observations where error is possible	Proportion of observations with error	Description of error
Diabetes	435	0.497	Reports having been diagnosed with diabetes in Wave 1 and never having been diagnosed with diabetes in Wave 2
TB	474	0.660	Reports having been diagnosed with TB in Wave 1 and never having been diagnosed with TB in Wave 2

(Adapted from Ardington and Gasealahwe 2012:26)

Direct observation

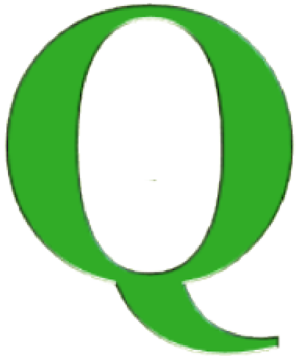
Collecting bio-specimens

TREATS project

Equipment

Logistic

Quality control



Safety

Cost

SANHNANES 2012: logistics problems…

Mobile (on wheels) clinics were a critical component in the planning of the survey. The NDoH had approved, and so informed the provincial heads of departments. The use of the mobile clinics would have enabled the clinical team to complete the clinical component of the fieldwork (physical examination, clinical tests, anthropometry and bloo sampling for biomarker analysis) in the close proximity of the residence of the participant who volunteered to take part in the survey. This arrangement would have ensured that th fieldwork would have been completed within the planned framework. However, mobile clinics were not made available for a number of province-specific reasons – this created the first major challenge to the survey.

In the absence of mobile clinics and the need to ensure participant safety and a good survey response rate, a number of innovative alternatives had to be considered in order to secure an environment that afforded privacy and feasibility for the clinical examination and testing of the participants. These alternatively created facilities included primary healthcare facilities with limited resources and space, school halls, church halls, hospice premises, city halls and community centres. Any other such premises that would create a sufficiently professional and private environment to examine and test the participants with the due respect were also used. For safety considerations, the participants had to be transported to these alternatively created facilities (using return home to 'clinic' transport).

(Shisana et al. 2014:49)

… and consequences

Table 3.1.3: Interview, physical examination and specimen response coverage by demographic characteristics, South Africa 2012

	Response profiles						
	Interviewed		Physical examination		Specimen collected		Total eligible
	%	n	%	n	%	n	n
Sex							
Male	91.9	11 273	40.1	4 917	26.6	3 270	12 271
Female	93.9	13 885	48.0	7 095	32.5	4 801	14 789

(Table from Shisana et al. 2014 - partial)

Participation rates, refusals,…

Bio-measures propensity by sponsorship.

		Very Unwilling	Somewhat Unwilling	Somewhat Willing	Very Willing
Height and Weight	Total	19.1%	19.1%	34.0%	27.7%
	Federal govt	20.5%	19.2%	34.9%	25.4%
	Health study	17.7%	19.1%	33.3%	29.9%
Blood pressure	Total	18.8%	17.8%	33.7%	29.6%
	Federal govt	20.2%	18.7%	34.0%	27.1%
	Health study	17.5%	17.0%	33.5%	32.0%
Finger-stick blood draw	Total	30.7%	24.3%	25.6%	19.5%
	Federal govt	32.6%	24.2%	25.6%	17.5%
	Health study	28.8%	24.3%	25.5%	21.4%
Waist circumference*	Total	19.9%	17.8%	34.3%	27.9%
	Federal govt	22.3%	18.5%	34.9%	24.3%
	Health study	17.6%	17.2%	33.8%	31.4%
Machine to measure blood pressure over three days**	Total	14.0%	14.6%	35.9%	35.5%
	Federal govt	16.8%	15.8%	35.7%	31.7%
	Health study	11.3%	13.5%	36.1%	39.1%
App on your phone**	Total	15.5%	16.7%	40.8%	27.0%
	Federal govt	20.9%	21.3%	37.4%	20.4%
	Health study	10.4%	12.3%	44.1%	33.3%

(Table adapted from Boyle et al. 2021)

Participation rates, refusals,…

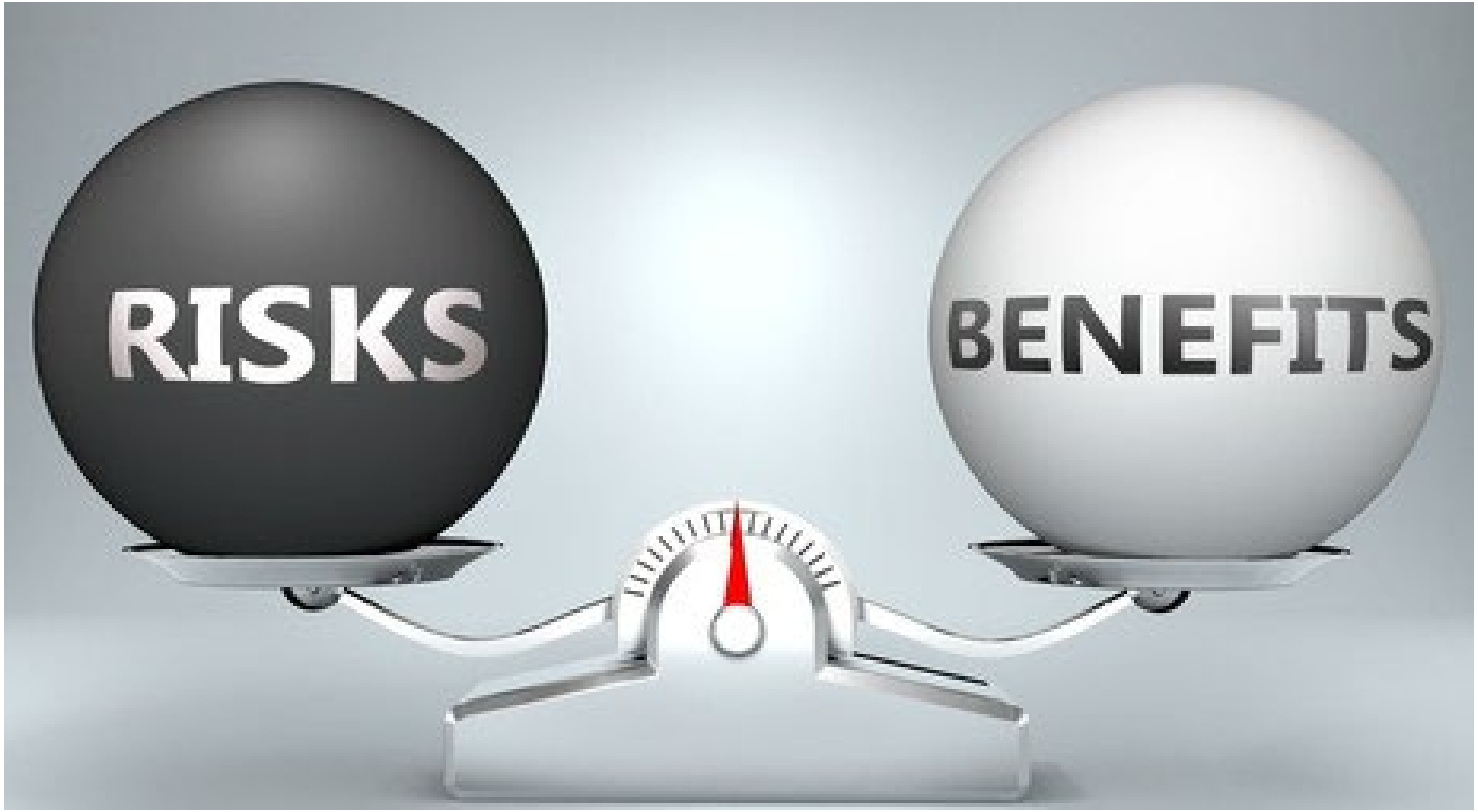
Table 4
Demographic Difference in Propensity for Bio-measures in terms of Composite Index based on Responses to Four Questions'

	Mean	N	Std. Dev.	SE of Mean
Gender*				
Male	10.71	883	3.94	.13
Female	10.29	1050	3.99	.12
Age***				
18–24	10.88	257	3.75	.23
25–34	11.02	345	3.78	.20
35–44	10.79	360	3.81	.20
45–54	10.37	379	4.01	.21
55–64	9.81	241	4.15	.27
65+	9.95	351	4.19	.22
Children under 18 in HH***				
Yes	11.08	800	3.82	.14
No	10.06	1133	4.02	.12
Children under 6 in HH***				
Yes	11.31	291	3.79	.22
No	10.34	1642	3.98	.10
Race/Ethnicity***				
White	10.32	1206	4.03	.12
Black	11.19	201	3.72	.26
Hispanic	11.01	365	3.87	.20
Other	9.68	161	3.81	.30

Differential response rate can introduce bias.

(Table from Boyle et al. 2021 - partial)

Ethics and the risk/benefit ratio



Summary questions

- Going beyond self-report: pros and cons
- Direct observation, medical procedures
- Collecting bio-specimens: ethical problems

References

Johnson, Timothy P. 2015. *Handbook of Health Survey Methods*. Vol. 565. John Wiley & Sons. Chapters 15 & 16.

Boyle, John, Lewis Berman, James Dayton, Ronaldo Iachan, Matt Jans, and Randy ZuWallack. 2021. 'Physical Measures and Biomarker Collection in Health Surveys: Propensity to Participate'. *Research in Social and Administrative Pharmacy* 17 (5): 921–29. https://doi.org/10.1016/j.sapharm.2020.07.025.

Pappas, Gregory, and Adnan A. Hyder. 2005. 'Exploring Ethical Considerations for the Use of Biological and Physiological Markers in Population-Based Surveys in Less Developed Countries'. *Globalization and Health* 1 (November): 16. https://doi.org/10.1186/1744-8603-1-16.

Other resources

Ardington, C., and B. Gasealahwe. 2012. *Health: Analysis of the NIDS Wave 1 and 2 Datasets.* Southern Africa Labour and Development Research Unit.

Thank you

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